

FIFA World Cup 2026

Prediction Systems Retrospective

Evolution, comparative performance, failure analysis and recommended successor architecture

Post-tournament professional review • 20 July 2026

Principal finding. Diamond v8.4 was the best empirically demonstrated system in the project's controlled comparison. EWE-26 v1.3.1 was the most coherent final architecture, but it was introduced too late and modified too adaptively to be declared the best-performing model on evidence alone.

Purpose and scope

This report reconstructs the development of the World Cup exact-score project from the available project conversations, workbooks and protocol packages. Match dates are treated as effective timestamps: a method is credited only with forecasts generated before the relevant match, while retrospective reruns are retained as development evidence but excluded from prospective performance claims.

The review addresses four questions: how the systems evolved; which strategy performed best; why the other systems failed; and which elements should be retained in a future immutable forecasting protocol. The available record is substantial but not a perfectly normalised prediction ledger. Accordingly, conclusions are classified as demonstrated, supported or provisional.

Executive conclusions

Best demonstrated system: Diamond v8.4. In the project's directly compared 12-match benchmark, it reportedly achieved 8 exact-score hits, 10 correct results and a mean absolute goal-distance of 0.583. It improved materially on v8.3 and v8.5 because it corrected draw inertia and favourite-score underexpansion without allowing phenotype overrides to dominate the underlying probability distribution.

Best conceptual system: EWE-26 v1.3.1. Its hierarchy—result first, goals only when durable, and tails only under convergent game-state evidence—was the clearest solution to the repeated Diamond failure modes. It also explicitly recognised elite-underdog winner flips and environmental cancellation of favourite goal floors.

Why these are not the same answer: Diamond v8.4 has comparative performance evidence. EWE-26 v1.3.1 has superior architecture but insufficient frozen-sample validation. Declaring EWE-26 the winner would confuse methodological maturity with demonstrated predictive performance.

Prospective ledger finding. Among the six exact tournament locks that could be reconstructed without ambiguity, no exact score was achieved. This confirms that the project's benchmark gains did not translate automatically into robust live exact-score performance and strengthens the case for an immutable ledger and frozen evaluation batches.

Primary cause of failure across the project: The system repeatedly concentrated sophistication in the final exact-score conversion layer while the underlying team-goal distribution, state dependence and uncertainty calibration remained less stable. This produced frequent switching among 1–0, 1–1, 2–0 and 2–1 without a commensurate change in the underlying evidence.

Primary governance failure: Versions were revised after small clusters of visible misses. These changes were often rational, but the tournament simultaneously served as operating environment, development set and validation set, creating unavoidable adaptive overfitting.

1. Evidence model and chronology

The project evolved through two broad families. Diamond began as a historically regularised, market-informed exact-score selector and later became a workbook-driven ensemble with rule-based overrides. EWE-26 then reorganised the problem as a hierarchical classification task: result regime first, scoring regime second, and exact score last.

Effective period	System	Core formulation	Assessment
Pre-tournament / early group stage	Diamond v6–v6.3	Historical score priors, market/xG estimates, line-up news and manual pathway corrections	Established ACP, clean-sheet fragility, dual-scorer and low-block concepts; still highly operator-dependent.
Early tournament development	Diamond v7–v8.3	Probability score, leaderboard score and evidence-block overrides	More structured, but hard overrides frequently overthought the baseline and draw selection remained excessive.
Benchmark-led revision	Diamond v8.4	Rule-governed draw release, favourite expansion and consolation-goal handling	Best measured balance of conservatism and selective score expansion.
Workbook integration phase	Diamond v8.5–v8.5.2	Player/coach registry, workbook-derived team metrics, separate public-search final lock	Operationally ambitious; deterministic phenotype overrides sometimes displaced correct high-margin outcomes.

Late group stage / knockout transition	EWE-26 v1.1–v1.2.2	Market-calibrated, result-first ensemble with DrawGate, CleanSheetGate, UGG and TailGate	Improved discipline and auditability, but early gate thresholds overprotected draws and suppressed some genuine consolation goals.
Late knockout phase	EWE-26 v1.3–v1.3.1	Durability gates, altitude/conditions, elite-underdog flips, low-block compression	Strongest final formulation; too little frozen prospective sample for a fair performance ranking.

2. System-by-system evolution

2.1 Diamond v6–v6.3: priors plus expert correction

The initial Diamond architecture was founded on a defensible empirical observation: World Cup scorelines are highly concentrated. The project baseline showed 1–0, 2–1, 1–1, 0–0, 0–1 and 2–0 as dominant historical score families. These priors reduced extreme predictions and provided a rational default when match-specific evidence was weak.

The v6.2 and v6.3 daily locks then added market odds, projected goals, expected line-ups and pathway judgements. Examples included keeping Germany–Curaçao at 3–0 because low-block and under-market signals closed the four-goal avalanche gate, and moving Netherlands–Japan from 1–1 to 2–1 because both sides had credible scoring routes but the Netherlands retained a slight personnel and market edge.

Failure mechanism. The approach was not sufficiently reproducible. “Credible route,” “kill power,” “clean-sheet fragility” and “cascade pressure” were useful concepts, but their weights were not estimated consistently. The operator could rationally justify either conserving or releasing a scoreline, which made the final lock vulnerable to narrative drift.

2.2 Diamond v7–v8.3: ensemble overrides

The next phase separated a conservative sanity score, a highest-probability score and a leaderboard-oriented score. The intended v8 rule was to retain the conservative baseline unless at least three independent evidence blocks supported an override. This was a meaningful advance because it distinguished pure probability optimisation from competition-point optimisation.

However, the broader archived 48-match stress-test run returned only 6 exact hits, 25 correct results and a 1.75 average total-goal error. Hard overrides were exact in approximately 11% of cases, providing direct evidence that the override layer was frequently adding noise rather than signal. The benchmark workbook itself stated the governing diagnostic correctly: if hard overrides did not beat the v6.1 baseline, the system was overthinking.

Failure mechanism. v8.3 retained excessive draw mass and allowed multiple correlated evidence blocks to masquerade as independent convergence. Market shape, projected xG and “goal-route” scoring were often derived from overlapping information. Three supporting blocks therefore did not necessarily represent three independent signals.

2.3 Diamond v8.4: calibrated rule governance

Diamond v8.4 was not a complete rebuild. Its advantage was restraint. It introduced more explicit rules for draw preservation, favourite-score expansion, underdog consolation goals and tail release, while keeping the underlying probability structure intact.

In the project's directly compared 12-match benchmark, v8.4 improved from the v8.3 baseline of 6 exact hits, 9 result hits and 1.0 mean goal-distance to 8 exact hits, 10 result hits and 0.583 mean goal-distance. This is the strongest comparative result documented within the project. It also outperformed v8.5, which returned 7 exact hits, 10 result hits and 0.917 mean goal-distance.

Why it worked. v8.4 addressed the most frequent errors at the correct level of complexity. It did not discard the baseline distribution; it regularised the conversion from that distribution to one leaderboard pick. It released draws only where the favourite edge was durable, expanded favourites only where the tail was supported, and treated an underdog goal as a thresholded event rather than an automatic consequence of non-zero attacking ability.

2.4 Diamond v8.5–v8.5.2: data architecture ahead of calibration

The v8.5 programme sought to transform Diamond into a standalone workbook execution: player registry, squad and coach information, team metrics, goal expectations and exact-score selection, followed by a separate final-lock public-search correction layer. This was the correct operational design for reproducibility.

The model nevertheless became more brittle. The project's comparison attributed the v8.5 regression to deterministic phenotype overrides that occasionally suppressed correct high-margin scores. In practical terms, a categorical label such as low block, conservative favourite or fragile underdog could dominate the continuous probability distribution and force an overly narrow score.

Failure mechanism. The input architecture was more sophisticated than the validated mapping from player-level variables to team-level scoring rates. The workbook therefore created an impression of mechanistic precision that exceeded the calibration of its coefficients. Better data plumbing did not automatically produce better score probabilities.

2.5 EWE-26 v1.1–v1.2.2: result-first restructuring

EWE-26 represented the most important conceptual break in the project. Instead of beginning with an exact score, it classified the match sequentially: outcome, goal environment, clean-sheet/BTTS state, tail eligibility and exact-score selection. The deterministic v1.2 protocol added fixed score-matrix construction, explicit CleanSheetGate and Disturbance scoring, revised confidence bands and mandatory consolation-goal suppression.

This structure directly targeted Diamond's recurring errors. Nevertheless, early prospective locks exposed threshold problems. Switzerland–Canada was held at 1–1 and subsequently diagnosed as draw overprotection. Bosnia–Qatar was held at 2–0 and subsequently diagnosed as both favourite-tail underexpansion and incorrect consolation-goal suppression. The new hierarchy was sound, but the initial gates were too conservative in marginal cases.

Failure mechanism. Binary or quasi-binary gates created discontinuities. A small change around a threshold could convert 1–1 to 2–1, or 3–1 to 3–0, even when the underlying probabilities were nearly continuous. The system improved auditability but did not yet fully calibrate the thresholds.

2.6 EWE-26 v1.3–v1.3.1: mature architecture

The final EWE formulation condensed the project's learning into five principles: result first; goals only if durable; tails only when game-state evidence is convergent; elite underdogs can flip the winner; and conditions can cancel favourite goal floors.

This was the strongest architecture because it separated distinct inferential questions and corrected prestige anchoring, false consolation goals, low-block compression and environmental effects. It was also appropriately sensitive to knockout-specific regimes, where advancement probability, regulation score and extra-time outcome must be separated.

Limitation. v1.3.1 cannot be ranked above v8.4 on demonstrated accuracy. It was introduced during the late knockout phase, after repeated exposure to tournament outcomes, and was followed by further aggressive/leaderboard adjustments. The available sample is too small and too adaptively selected for a fair empirical comparison.

3. Comparative assessment

System	Observed performance	Reproducibility	Overfitting risk	Overall judgement
Diamond v6–v6.3	Moderate	Low–moderate	High	Useful foundation; too narrative-dependent.
Diamond v8.3	Weak–moderate	Moderate	Moderate	Structured but overrode too often and overpredicted draws.
Diamond v8.4	Strongest demonstrated	High	Moderate	Best empirical balance; recommended benchmark champion.
Diamond v8.5–v8.5.2	Moderate	High in workflow, lower in parameter validity	Moderate	Excellent pipeline ambition; phenotype rules became overdetermining.
EWE-26 v1.2–v1.2.2	Promising but mixed	High	Low–moderate	Superior hierarchy; gate thresholds initially too conservative.
EWE-26 v1.3.1	Insufficient sample	High	Low in principle, high due to late adaptation	Best successor architecture, not yet validated champion.

4. Cross-system failure analysis

- **Outcome and score were insufficiently separated.** Earlier systems often allowed a preferred exact score to determine the implied result. EWE correctly reversed this ordering.
- **Correlated evidence was counted as convergence.** Odds, xG proxies, market totals and route ratings frequently originated from the same market information. The apparent evidence count therefore overstated independence.

- **Draw preservation became a default rather than a calibrated probability.** Balanced routes were repeatedly converted into 1–1 even when the favourite-win mass across 1–0, 2–0 and 2–1 exceeded the draw score.
- **Consolation goals were alternately automatic or over-suppressed.** Diamond often granted the underdog a goal because one route existed; early EWE sometimes removed it unless several routes converged. Both errors arose from hard thresholding.
- **Favourite tails were handled inconsistently.** Low-block logic sometimes prevented deserved 3–0 or 4–0 predictions; cascade narratives sometimes opened tails without sufficient evidence.
- **Player-level detail was not matched by validated aggregation.** The v8.5 workbook improved data coverage, but player and coach variables were not yet linked to team lambdas through a sufficiently estimated model.
- **Tournament regimes were incompletely separated.** Group-stage incentives, knockout regulation, extra time, penalties and third-place playoffs have different scoring distributions. A universal score-conversion rule was inappropriate.
- **Post-match revision occurred too rapidly.** Small clusters of errors triggered structural changes. This increased responsiveness but reduced the ability to distinguish genuine improvement from tournament-specific overfitting.

5. Determination of the best strategy

Best demonstrated strategy	Diamond v8.4
Best strategy for future development	EWE-26 v1.3.1 architecture, recalibrated and frozen before testing

Diamond v8.4 should be regarded as the project champion because it is the only late-stage system with a direct comparative benchmark showing simultaneous improvement in exact-score hits, result hits and goal-distance error. Its success came from controlled correction rather than wholesale reinterpretation of the match.

EWE-26 v1.3.1 should nevertheless form the basis of the next model. Its inferential hierarchy is more statistically coherent and better aligned with the failure analysis. The correct synthesis is therefore not to revert permanently to Diamond v8.4, but to place v8.4-style conservative regularisation inside an EWE result-first architecture.

6. Recommended successor protocol

The recommended post-tournament successor should be frozen before any new competition and should contain the following layers in order:

1. **Outcome distribution:** Estimate home/neutral win, draw and away win probabilities. Do not select an exact score at this stage.
2. **Continuous scoring distribution:** Estimate team goal rates and their covariance using market, team, player, tactical and environmental variables. Preserve the full score matrix.

- 3. Regime classification:** Classify group-stage, knockout regulation, extra time and third-place fixtures separately.
- 4. Durability weighting:** Represent clean sheets, underdog goals and favourite tails as continuous probability modifiers rather than binary gates.
- 5. Conservative exact-score selector:** Use a v8.4-style regularised selector that defaults to the modal score and permits deviations only when the posterior probability gain or leaderboard utility is explicit.
- 6. Separate leaderboard layer:** Publish probability-first and aggressive picks as distinct outputs. Never silently substitute one objective for the other.
- 7. Immutable audit ledger:** Record model version, timestamp, information cutoff, probability matrix, prelock, final lock, override reason and actual 90-minute result.
- 8. Frozen evaluation:** No rule change until a predeclared batch is complete. Evaluate exact accuracy, result accuracy, mean score distance, ranked probability score, log loss and calibration.

7. Match-level prediction audit and effectiveness scoring

The table below contains the prospective tournament predictions that could be recovered with a verifiable pre-match timestamp, an explicit exact-score lock and a confirmed final score. Match kickoff chronology is used as the version boundary. Predictions mentioned only as retrospective reruns, incomplete conversational references or unrecorded “final locks” are excluded from numerical scoring rather than reconstructed from memory.

Effectiveness scale. Exact score = 10.0. A correct result but incorrect score is scored as $7.5 - 0.75 \times \text{absolute goal error}$, with a minimum of 4.0. An incorrect result is scored as $4.0 - 0.75 \times \text{absolute goal error}$, with a minimum of 0. Absolute goal error is $|\text{predicted home goals} - \text{actual home goals}| + |\text{predicted away goals} - \text{actual away goals}|$. This score measures closeness of the submitted single score; it is not a substitute for probabilistic log loss or calibration.

Date	Match	System / lock	Prediction	Actual	Result	GAE	Eff. /10
14 Jun	Haiti–Scotland	Diamond v6.3 final	1–2	0–1	Correct	2	6.0
14 Jun	Australia–Türkiye	Diamond v6.3 final	0–1	2–0	Incorrect	3	1.8
14 Jun	Germany–Curaçao	Diamond v6.3 final	3–0	7–1	Correct	5	4.0
14 Jun	Netherlands–Japan	Diamond v6.3 final	2–1	2–2	Incorrect	1	3.3
24 Jun	Switzerland–Canada	EWE-26 v1.2 prelock	1–1	2–1	Incorrect	1	3.3
24 Jun	Bosnia–Qatar	EWE-26 v1.2 prelock	2–0	3–1	Correct	2	6.0

Match-specific diagnoses.

- 1. Haiti–Scotland:** Winner identified, but Haiti was granted a non-durable consolation goal and Scotland’s ceiling was overstated.
- 2. Australia–Türkiye:** Market-favourite anchoring dominated Australia’s home and set-piece routes; both the result direction and clean-sheet assignment failed.

3. **Germany–Curaçao:** The favourite was identified, but ACP/tail closure was severely over-conservative and the Curaçao goal route was wrongly suppressed.
4. **Netherlands–Japan:** The dual-scorer correction was directionally useful, but forced favourite release converted the correct draw family into a false narrow win.
5. **Switzerland–Canada:** DrawGate overprotected equilibrium despite a sufficient Switzerland winner edge; the score distance was small but the primary result call failed.
6. **Bosnia–Qatar:** The winner and margin family were correct, but TailGate underexpanded Bosnia and mandatory consolation suppression removed Qatar’s realised goal.

Recoverable prospective sample summary. Diamond v6.3: n=4, exact scores 0/4, correct results 2/4, mean absolute goal error 2.75, mean effectiveness 3.75/10. EWE-26 v1.2: n=2, exact scores 0/2, correct results 1/2, mean absolute goal error 1.50, mean effectiveness 4.63/10. The apparent EWE advantage is descriptive only because both samples are small, non-random and drawn from different tournament phases.

7.1 Direct system comparison available within the project

The strongest head-to-head evidence is the project’s frozen 12-match comparison of Diamond v8.3, v8.4 and v8.5. These data are retained separately from the tournament ledger because they are a controlled benchmark rather than a complete chronology of prospective World Cup locks. A composite effectiveness index is provided solely to facilitate comparison: 50% exact-score rate, 30% result rate and 20% goal-distance quality, where goal-distance quality = $\max[0, 1 - \text{mean absolute goal error}/3]$.

System	n	Exact	Result	Mean GAE	Composite /100	Interpretation
Diamond v8.3	12	6/12	9/12	1.000	65.8	Too many draw-preserving and correlated-evidence overrides.
Diamond v8.4	12	8/12	10/12	0.583	79.4	Best balance: conservative baseline retained, selective draw/tail release.
Diamond v8.5	12	7/12	10/12	0.917	72.6	Player/workbook architecture improved, but phenotype overrides became overdetermining.

Interpretation. Diamond v8.4 remains the best empirically demonstrated system because it improved exact-score accuracy, preserved the same result accuracy as v8.5 and produced the lowest score-distance error. Its advantage was not greater complexity, but controlled restraint: it changed the baseline only when the evidence for draw release, favourite expansion or an underdog goal was sufficiently durable. EWE-26 v1.3.1 remains the preferred successor architecture, but no equally sized frozen prospective comparison is available.

7.2 Coverage limitations of the reconstructed ledger

The project history confirms that additional matches were analysed under Diamond v8.4/v8.5.2 and EWE-26 v1.3/v1.3.1, including late group-stage and knockout fixtures. However, the currently recoverable record does not consistently expose the exact submitted score, whether it was a prelock or final lock, and the model version in one auditable entry. These matches are therefore part of the

qualitative systems history but are not assigned effectiveness scores in this edition. This conservative exclusion prevents retrospective selection of whichever conversational alternative happened to be closest to the result.

8. Evidentiary limitations

The project conversations provide chronological version boundaries and numerous prospective locks, but they do not constitute a single machine-readable ledger of every match. Some matches have both prelock and final lock records; others have only conversational summaries; some evaluations were retrospective blind-style reruns. The report therefore avoids presenting a tournament-wide exact-score percentage that cannot be independently reconstructed from the currently normalised evidence.

The 12-match v8.3/v8.4/v8.5 comparison is treated as the strongest direct comparative evidence available. The separate 48-match archived v8 run is informative regarding override failure, but it relied partly on proxy totals and algorithmically reconstructed inputs, so it is not equivalent to a fully prospective tournament test.

9. Final judgement

The project's best-performing strategy was Diamond v8.4 because it made the smallest effective correction to the baseline and produced the strongest documented comparative metrics. The project's most scientifically defensible endpoint was EWE-26 v1.3.1 because it reorganised prediction around outcome, durability and game-state convergence. The optimal future system is a frozen hybrid: EWE's hierarchy, v8.4's regularisation, continuous rather than binary gates, and a strict prospective audit ledger.

Project evidence consulted

- Diamond v6 Baseline Summary: historical exact-score priors for World Cup and broader international matches.
- Diamond v6.2 and v6.3 Daily Lock Sheets, 14 June 2026: market, line-up, projected-goal and pathway-based prospective selections.
- Diamond v8 Stress-Test Workbook and Run 1 archived/proxy input notes: blind-lock protocol, evidence-block overrides and 48-match metrics.
- Diamond WC2026 Day-Before Structured Dataset: fixture, squad, Elo, role, injury and expected-line-up input architecture.
- Project conversation chronology: Diamond v8.4/v8.5.2 benchmark comparisons; EWE-26 v1.1–v1.3.1 protocol changes; prospective locks and post-match diagnoses.